WADF105 – Network Security Fundamentals

Practical Laboratory 1: Two VM Network Commissioning and Connectivity Verification

Student Name: Abdul Fatawu Umar
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1. Introduction

In this laboratory, I configured a small protected network using two virtual machines: **icdfa-nslab-firewall-v1** running OPNsense and **icdfa-nslab-client-v1** running Ubuntu Linux. The purpose of the exercise was to understand the roles of firewall WAN and LAN interfaces, verify IPv4 addressing and routing, test network connectivity, and observe network traffic using Wireshark.

The lab helped me gain practical experience with firewall deployment, DHCP addressing, default gateway configuration, packet analysis, and troubleshooting connectivity issues within a virtualized environment.

2. Objectives

The objectives of this laboratory were to:

- Configure a two-VM VirtualBox network environment.
- Understand the functions of WAN and LAN interfaces on a firewall.
- Obtain and verify valid IPv4 addressing information.
- Verify gateway configuration and routing.
- Test local network connectivity and internet access.
- Access the OPNsense web management interface.
- Capture and analyse ARP, ICMP, and DNS traffic using Wireshark.

3. Laboratory Environment

The laboratory consisted of the following virtual machines:

Virtual Machine	Function
icdfa-nslab-firewall-v1	OPNsense Firewall
icdfa-nslab-client-v1	Ubuntu Client

The OPNsense firewall was configured to provide routing and DHCP services for the Ubuntu client. VirtualBox NAT was used to provide internet connectivity to the firewall WAN interface, while an Internal Network named **ICDFA-LAN** connected the Ubuntu client to the firewall LAN interface.

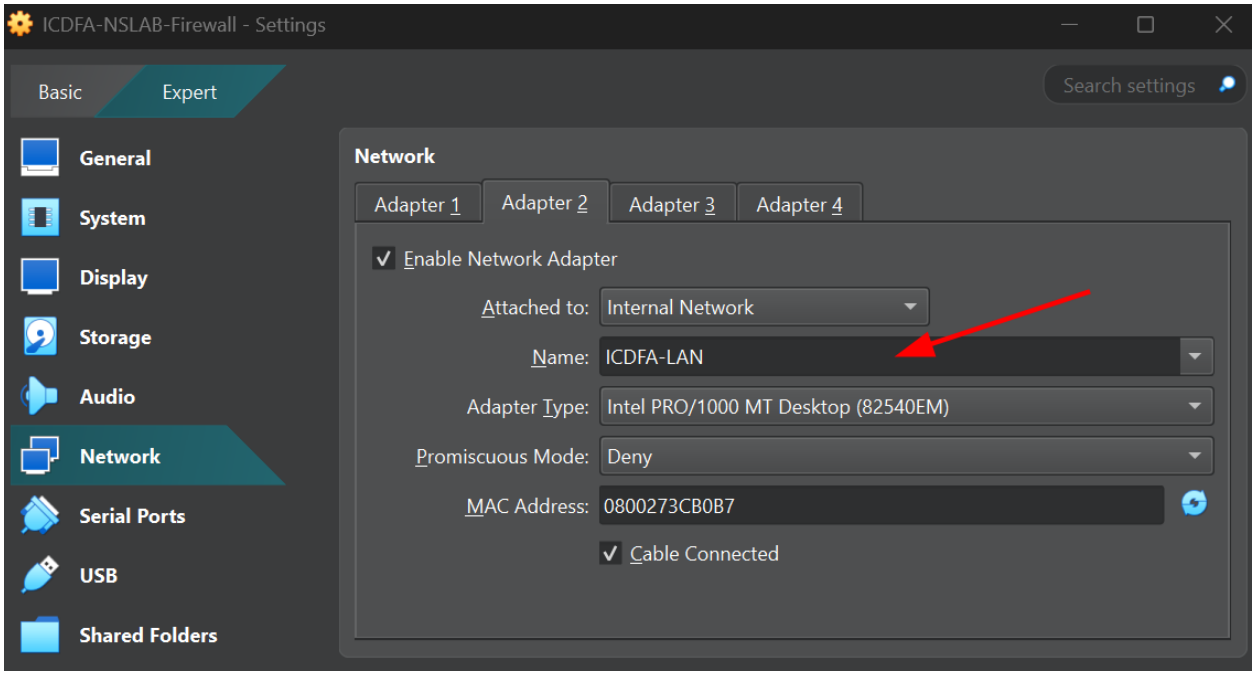
4. VirtualBox Network Configuration

Firewall Configuration

I opened the network settings of **icdfa-nslab-firewall-v1** and configured:

- Adapter 1 as NAT
- Adapter 2 as Internal Network
- Internal Network Name: ICDFA-LAN

Unused adapters were disabled as required by the laboratory instructions.



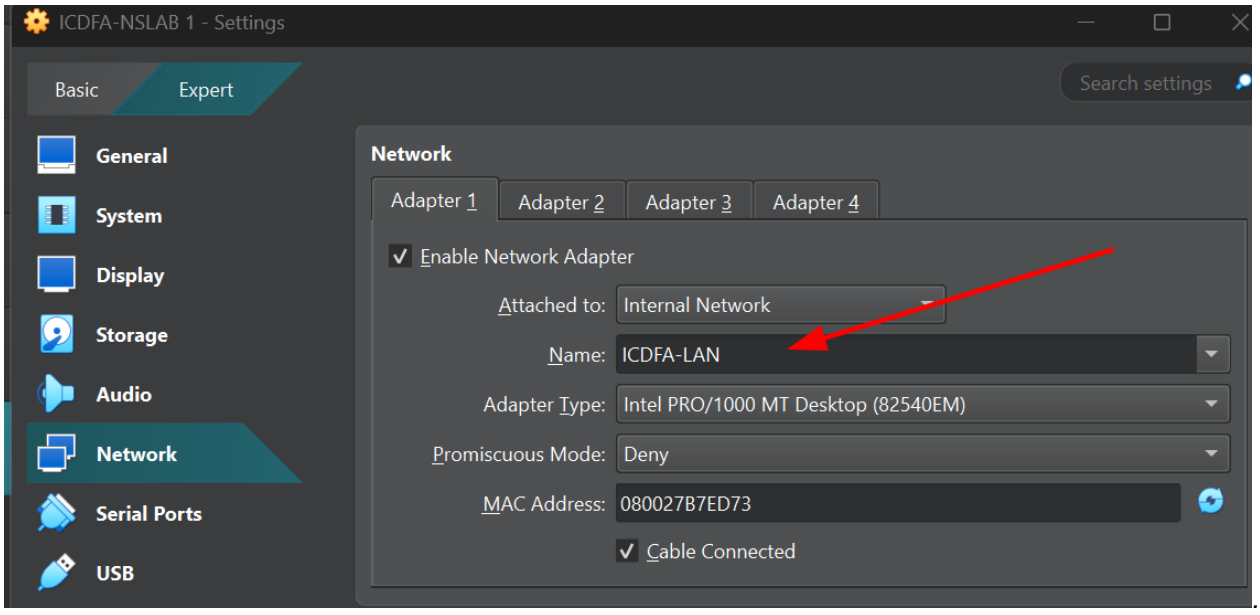
Caption: OPNsense firewall network configuration showing NAT and ICDFA-LAN adapters.

Ubuntu Configuration

I configured **icdfa-nslab-client-v1** with:

- Adapter 1 as Internal Network
- Network Name: ICDFA-LAN

All additional adapters were disabled.



Caption: Ubuntu network settings configured to connect to ICDFA-LAN.

5. OPNsense Interface Verification

After starting the firewall VM, I verified that the LAN interface was configured with the address:

10.10.10.1/24

I confirmed that DHCP was enabled on the LAN interface to automatically assign addresses to client systems.

The OPNsense console displayed the configured interfaces and their corresponding IP addresses.

6. Ubuntu Network Verification

After starting the Ubuntu client, I opened a terminal and used the following command:

ip -4 -br address

The output showed that the Ubuntu client successfully received an IPv4 address from the DHCP server.

My assigned address was:

10.10.10.105/24

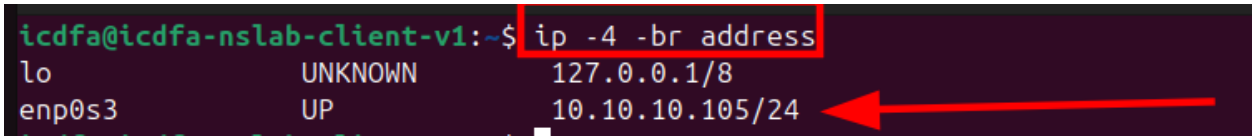
I then verified the routing table using:

ip route

The default gateway was configured as:

10.10.10.1

which corresponds to the OPNsense LAN interface.



Caption: Ubuntu IPv4 configuration and routing information.

7. Connectivity Testing

To verify connectivity, I performed several network tests.

Test 1: Communication with the Firewall

I ran:

ping -c 4 10.10.10.1

The firewall responded successfully to all ping requests, confirming communication between the Ubuntu client and the OPNsense LAN interface.

Test 2: Internet Connectivity

I attempted to test internet connectivity using:

ping -c 4 1.1.1.1

The test resulted in 100% packet loss.

During troubleshooting, I discovered that the Ubuntu client could communicate with the firewall successfully but could not reach external networks. This indicated that the LAN configuration was functioning correctly while the issue was likely related to the firewall WAN configuration.

Test 3: DNS Activity

I executed:

getent hosts opnsense.org

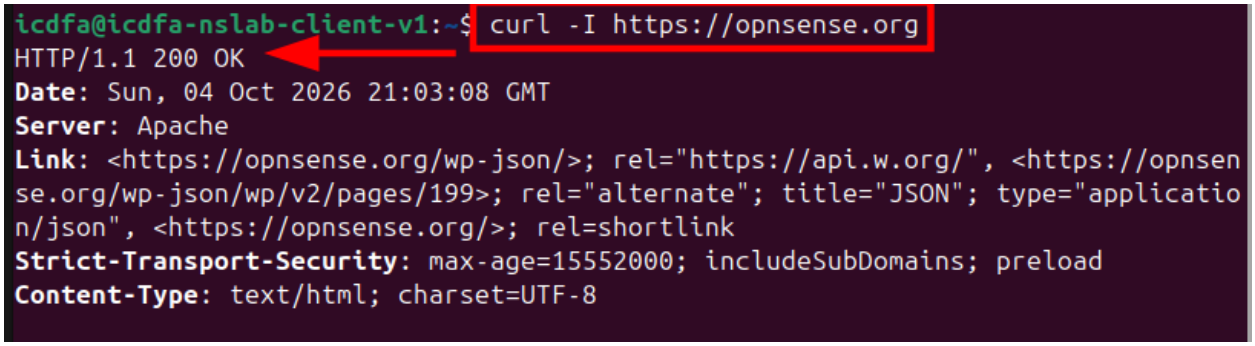
to generate DNS traffic for packet analysis in Wireshark.

Test 4: Web Connectivity

I ran:

curl -I https://opnsense.org

as instructed, to test web communication.



Caption: Connectivity testing performed from the Ubuntu client.

8. Wireshark Packet Analysis

I used Wireshark to observe network traffic between the Ubuntu client and the OPNsense firewall.

I started packet capture on the active Ubuntu network interface and generated traffic by pinging the firewall and performing DNS lookups.

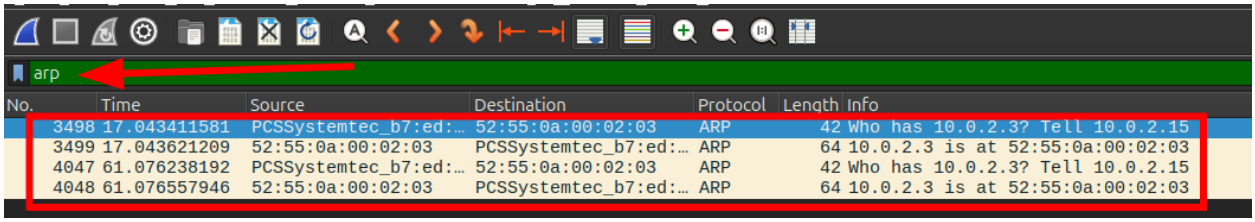
ARP Analysis

I applied the filter:

arp

and observed ARP request and response packets.

The ARP exchange allowed Ubuntu to determine the MAC address associated with the firewall IP address.



Caption: ARP request and response traffic captured in Wireshark.

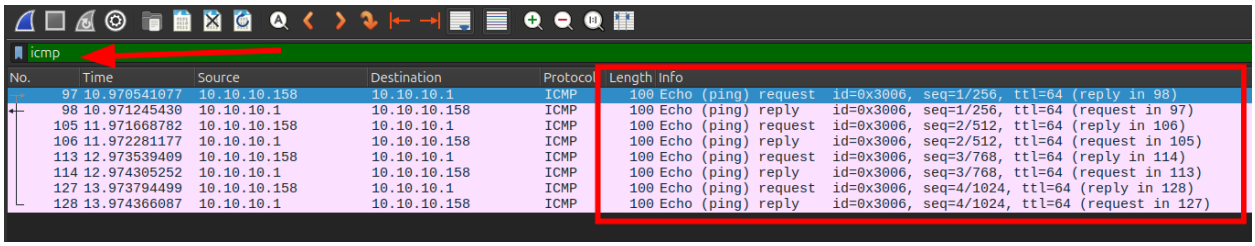
ICMP Analysis

I applied the filter:

icmp

and observed Echo Request and Echo Reply packets generated by the ping test.

These packets demonstrated successful communication between Ubuntu and the firewall.



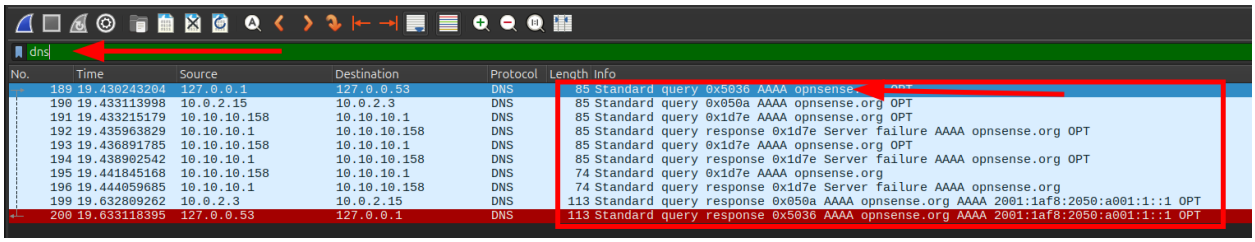
Caption: ICMP Echo Requests and Echo Replies captured during ping testing.

DNS Analysis

I applied the filter:

dns

and observed DNS query and response traffic generated during the hostname lookup.



Caption: DNS query and response packets captured in Wireshark.

9. Why Ubuntu Uses 10.10.10.1 as the Default Gateway

The Ubuntu client uses **10.10.10.1** as its default gateway because the OPNsense firewall acts as the router for the ICDFA-LAN network. Whenever Ubuntu needs to communicate with systems outside its local subnet, it forwards traffic to the firewall through the default gateway. The firewall then determines the appropriate route and forwards the traffic toward its destination. Without a default gateway, Ubuntu would only be able to communicate with devices on the local network.

10. Answers to Completion Questions

29. What is the difference between the OPNsense WAN and LAN interfaces?

The WAN interface connects the firewall to external networks such as the internet, while the LAN interface connects internal devices within the protected network.

30. Why must both internal adapters use the same VirtualBox network name?

VirtualBox only allows communication between virtual machines connected to the same internal network. Using different names creates separate virtual networks and prevents communication.

31. What information does the default route provide to Ubuntu?

The default route tells Ubuntu where to send traffic when the destination is outside the local network. In this lab, the default route points to the OPNsense firewall at 10.10.10.1.

32. Which packet exchange allows Ubuntu to learn the firewall MAC address?

The Address Resolution Protocol (ARP) request and ARP reply exchange allows Ubuntu to learn the MAC address associated with the firewall IP address.

33. Why does a successful ping to 1.1.1.1 not automatically prove that DNS is working?

A ping to 1.1.1.1 uses a direct IP address and does not require DNS name resolution. DNS functionality is only confirmed when a hostname such as opnsense.org can be resolved successfully.

Conclusion

In this laboratory, I successfully configured a two-VM network consisting of an OPNsense firewall and an Ubuntu client. I verified IP addressing, routing information, default gateway configuration, and communication between the Ubuntu client and the firewall. I also captured and analysed ARP, ICMP, and DNS traffic using Wireshark, which improved my understanding of how network devices discover one another and exchange information on a network.

Although internet connectivity through the firewall WAN interface required further troubleshooting, the laboratory successfully demonstrated the fundamental concepts of network addressing, routing, packet analysis, DHCP operation, and firewall-based network segmentation.

Ethics Statement: I conducted this laboratory entirely within the authorised ICDFA training environment and used the systems solely for educational and learning purposes.